

Rig: NaborsX10

API No.: 33 – 06 – 05180

Oil Field: ALKALI CREEK

WBS / Network #: 105-05030.04.01/ 5016166

Permit/WF#: 39894

Objective: MB Lowstand

KB Height, ft: 32.0

Co., State: MOUNTRAIL, ND

Surf Location: Sec. 7 154 N 94 W

BH Location: 160 ft FNL & 590 ft FWL, SECTION 05, T154N, R94W

Hard Line: 150'FNL/100'FSL

Latitude: 48° 10' 15.55"

Longitude: -102° 49' 3.91"

GL Elevation: 2,198 ft, graded, above MSL

Notes & Deviation

- Middle Bakken Lateral

- TD well at least 160ft from North Section Line

Formation Tops TVD	Planned Wellpath	Hole Size/Depth	Mud Properties	Offset Hazards and Risks	Proposed BHA	Casing / Completions Program	Cement Program
Pierre 1,835 Greenhorn 4,263 Mowry 4,641 Swift 5,435 Rierdon 5,862 Piper_Lime 6,159 Top_Dunham_Salt 6,399 Spearfish 6,446 Top_Pine_Salt 6,744 Top_Opeche_Salt 6,976 Amsden 7,353 Tyler 7,536 Kibbey_Lime 8,120 Charles 8,290 Ratcliffe 8,908 Top_Last_Salt 8,992 Base_Last_Salt 8,941 Rival 9,178 Lodgepole 9,832 Bakken 10,578 Middle_Bakken 10,600 Base_Mid_Bakken 10,662 Three_Forks 10,718	KB, ft 32 7.6 Tangent, deg 10,095 TOL, MD 10,195 KOP, MD 12.5 deg/100 ft Build	16 in. Surf to 80 ft MD/TVD 13-1/2 in. Surf to 2,029 ft MD 8-3/4 in. 1,885 ft MD/TVD to 10,968 ft MD 10,634 ft TVD 6 in. 10,968 ft MD to 21,279 ft MD 10,579 ft TVD	8-3/4" Intermediate Section: Halliburton Integrate Bara-Xcel OBM Shoe to KOP: 9.5 - 9.8 ppg TD KOP w/ 9.5 - 9.8 ppg Enter UBS w/ 10.0 - 10.2 ppg* Land Curve w/ 10.2 ppg* HTHP Filtrate: < 4ml OWR: 60/40 Tau 0: 3 - 6 Utilize centrifuge as needed to keep LGS below 7.5%. 6" Lateral Section: Salt Water MW: 9.0-10.0 ppg LGS: < 5% pH: 8 - 9 -Pump sweeps as needed using Drillchem HE Lube	- One active offset SWD well 1.3 miles away (EN-MOUNTRAIL 53-102 SWD1 operated by Hess). - One active offset SWD well 2.5 miles away (NELSON 14-33H operated by Tallgrass Water North Dakota). - One new offset SWD well 2.5 miles away (NELSON SWD 2 operated by Tallgrass Water North Dakota). - One active offset SWD well 4.4 miles away (Hofflund 3 operated by Rim Operating). - One active offset SWD well 5.4 miles away (KRENTZ SWD #1 operated by PRAIRIE DISPOSAL).	8-3/4" Vertical BHA (1st BHA): 8-3/4" TKC66 TKCBQ3 7-1/4" ERT NOV 1.5°, 0.23 RPG, 7/8, 5.7 Stg; 1,185 psi max Differential P. at 180 F MWD Gordon Filter Sub 5" NC50 HWDP (21 jts) 5" NC50 DP 8-3/4" Vertical BHA (2nd BHA): 8-3/4" NOV TKC66 BC2 bit 7-1/4" ERT NOV 1.5°, 0.23 RPG, 7/8, 5.7 Stg; 1,185 psi max Differential P. At 180 F MWD Gordon Filter Sub 5" NC50 HWDP (21 jts) 5" NC50 DP 8-3/4" Curve BHA: 8-3/4" ULTERRA CF516 Single row bit 6-3/4" ERT NOV 2.38°, 0.47 RPG, 4/5, 7.0 Stg; 1,290 psi max Differential P. NMDC/MWD Gordon Filter Sub 5" NC50 HWDP (21 jts) 5" NC50 DP 6" Lateral BHA: 6" BAKER D405W or NOV TKC63 bit 5.125" Baker Duramax RS 1.6° MM, or 5.25" NOV 1.5°, 0.60 rpg, 6/7, 7.8 Stg MM NMDC/MWD Gordon X/O Filter Sub 4" DP (+/- 120 jts) 5" Agitator/XRV	9-5/8 in. Surface 9-5/8 in. 36 lb/ft J-55 STC Preset at 2,029 ft ref GL Note: Got back 68 bbl of cement returns to surface. 7 in. Production 7 in. 32 lb/ft P-110RY BTC Surface to 6,300 ft 7 in. 32 lb/ft P-110EC BTC 6,300 ft to TD 4-1/2 in. Liner 13.5 lb/ft P110 DWCC - Curve 11.6 lb/ft P110 BT&C - Lateral 4-1/2 in. BTC Frac String	9-5/8 in. Cement Lead 60 % Exces Type 1L 12.0 ppg 2.52 cu ft/sack Tail: 20 % Exces Type 1L 13.0 ppg 1.98 cu ft/sack Cement Service Provider: Halliburton (13/14.7Elasticem) 7 in. Cement Spacer: 50 bbl Gelled Salt Water 60 bbl Tuned Spacer Top of Lead Cement, ft 2,029 13.0 ppg 1.76 cu ft/sack 515 sks ElastiCem Top of Tail Cement, ft: 7,500 14.7 ppg 1.96 cu ft/sack 275 sks ElastiCem 406 bbl Approx. Displacement 4-1/2 in. Cement 14.3 ppg 1.56 cu ft/sack 687 sks Class G

22 deg Azimuth to land 7 in.22 deg Lateral Drilling AzimuthFormation Dip ± 90.55 S to N

ORGANIZATION APPROVALS

Tom Richardson
Drilling Team Lead

Jim Stover
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Isabel Leon
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